

EC 111: Principles of Microeconomics

Lecture Notes, Chapter 5: Elasticity and Its Application

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5 Elasticity and Its Application

Chapter 4 told us which way prices and quantities move when something changes in a market. Supply falls, the price rises and the quantity falls; demand rises, both rise. It did not tell us *how much*. That missing number decides almost every practical question: whether a tax mostly cuts consumption or mostly raises revenue, whether a record harvest is good or bad news for farmers, whether raising your own price earns you more or less. This chapter builds the measure, called elasticity, and then uses it.

5.1 A market without information: fish in Kerala

We start with the paper you read for today: Robert Jensen, “The Digital Provide: Information (Technology), Market Performance, and Welfare in the South Indian Fisheries Sector,” *Quarterly Journal of Economics* 122(3), 2007. It is a Chapter 4 story with a Chapter 5 punchline.

5.1.1 The setting

Kerala, on the southwest coast of India, has a large sardine fishery. Fish is a staple: over 70 percent of adults eat it at least once a day. Fishermen go out at night and sell the catch in the morning at beach markets along the coast. Three facts shape the market.

- The markets are short. Most were open only from 5:00 to 8:00 AM.
- A boat can reach only one market a day. Fuel is expensive and the market closes early.
- Fish does not keep. There is almost no storage, and no way to move fish along the coast by road once it has landed.

The result is that a fisherman chooses his beach *before* he knows what it pays. At sea he cannot see any price.

Example 5.1 (One morning: Tuesday, January 14, 1997). Jensen records the 7:45–8:00 AM sardine price in fifteen markets that morning. At Badagara the price was **zero**: eleven boats arrived to find the buyers gone and dumped their catch into the sea. Chombala and Quilandi, both within 15 kilometers, paid 9.9 and 9.8 rupees per kilogram, and buyers there went home without fish. The average boat that day carried 381 kg, and fuel for a 15 km trip cost about 205 rupees; a boat that landed at Badagara forwent as much as 3,400 rupees in profit.

5.1.2 What the phones did

Between 1997 and 2001, mobile phone service reached the Kerala coast, region by region. Once fishermen had phones they called ahead from the boat and chose the beach that was paying. Figure 1 shows what happened to prices in the first region to get coverage.

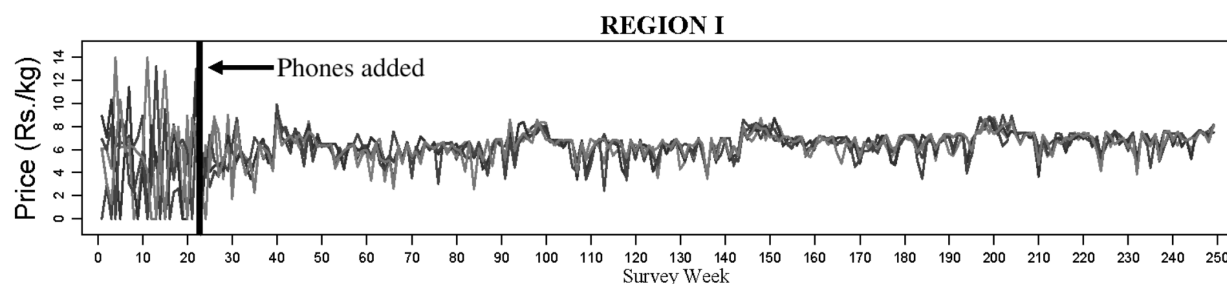


Figure 1: Sardine prices in the markets of Region I, one line per market, average 7:30–8:00 AM price in 2001 rupees per kilogram. Mobile phone service arrived in survey week 23. Source: Jensen (2007), Figure IV.

Before week 23, on the same morning, one market pays 14 rupees and another pays nothing. After week 23 the lines sit almost on top of each other. The same thing happened in Region II when phones arrived there in week 98, and in Region III in week 198. Because each region's prices changed at its own date and not at a common one, the change is hard to attribute to anything but the phones. Economists call a setting like this a *natural experiment*.

	Before phones	After phones
Price dispersion across markets (std. dev. / mean)	60–70%	15% or less
Market pairs violating the law of one price	50–60%	almost none
Share of the daily catch wasted	5–8%	0
Fishermen's profits		+8%
Consumer price		–4%

Table 1: Kerala before and after mobile phones. Source: Jensen (2007).

Key point

Fishermen earned more *and* consumers paid less. No one caught an extra fish. The gain came from two places: the 5–8 percent of the catch that used to be dumped was now sold, and fish now landed where buyers wanted it. This is Principle 6 from Chapter 1, measured. Prices coordinate a market only when people can see them.

5.1.3 Why the price swings were so large

Look again at the left side of Figure 1. Prices at a single market did not drift by a rupee or two from week to week. They jumped from 14 to 0. Supply and demand explain why (Figure 2).

On a given morning at a given beach, the catch has already been caught. Supply is fixed at whatever the boats bring in: a *vertical* line. Demand at one beach is steep: the local buyers are few, the market closes at 8:00, and fish that is not sold is worth nothing. When a few extra boats happen to land, supply moves right along a steep demand curve, and the price does not fall a little. It collapses.

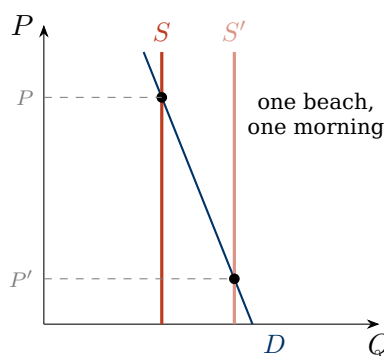


Figure 2: One market, one morning. Supply is the catch already landed. With steep local demand, a modest increase in the catch (S to S') moves the price a long way.

How far the price moves when quantity moves depends on how steep, or more precisely how *inelastic*, demand is. The rest of the chapter makes that idea exact.

5.2 The price elasticity of demand**5.2.1 The question**

Suppose you build websites for local businesses. You charge \$200 a site and sell 12 a month. Your costs are rising, including the opportunity cost of your own time, and you are thinking about charging \$250. The law of demand says you will sell fewer sites. It does not say how many fewer, and it does not say whether your revenue will go up or down. For that you need a number.

Definition 5.1 (Elasticity). **Elasticity** measures how much the quantity demanded or supplied responds to a change in one of its determinants.

Definition 5.2 (Price elasticity of demand). The **price elasticity of demand** measures how much the quantity demanded of a good responds to a change in its own price:

$$E_d = \frac{\text{percentage change in quantity demanded}}{\text{percentage change in price}}.$$

Along a demand curve price and quantity move in opposite directions, so this ratio is always negative. The convention is to drop the minus sign and report price elasticities of demand as positive numbers; a larger number means a more responsive quantity.

Elasticity uses percentages, not units. The number is the same whether fish is priced in rupees or dollars, per kilogram or per ton. That is what lets us compare the responsiveness of demand for sardines, websites, and insulin on one scale.

5.2.2 Computing it: the midpoint method

Figure 3 shows your demand as two points: A (\$200, 12 sites) and B (\$250, 8 sites). The obvious way to compute percentage changes is to divide by the starting value. It gives two different answers depending on the direction you go.

- From A to B: the price rises $50/200 = 25\%$ and quantity falls $4/12 = 33\%$, so $E_d = 33/25 = 1.33$.
- From B to A: the price falls $50/250 = 20\%$ and quantity rises $4/8 = 50\%$, so $E_d = 50/20 = 2.50$.

Same two points, two answers. The base you divide by is doing the damage.

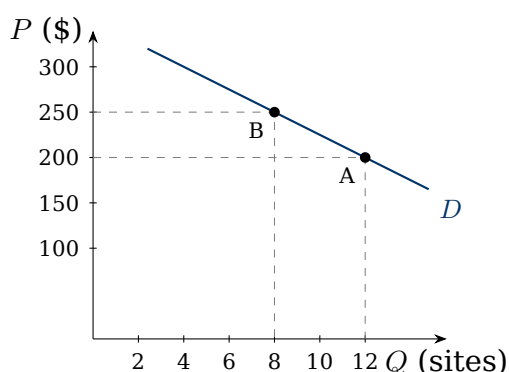


Figure 3: Demand for your websites. A: \$200, 12 sites a month. B: \$250, 8 sites.

Definition 5.3 (Midpoint method). Compute each percentage change relative to the average of the two values:

$$\% \Delta Q = \frac{Q_2 - Q_1}{(Q_1 + Q_2)/2}, \quad \% \Delta P = \frac{P_2 - P_1}{(P_1 + P_2)/2}, \quad E_d = \left| \frac{\% \Delta Q}{\% \Delta P} \right|.$$

Example 5.2 (Your websites, by the midpoint method). Between A and B the average quantity is 10 and the average price is \$225. So $\% \Delta Q = -4/10 = -40\%$ and $\% \Delta P = 50/225 = 22.2\%$, and

$$E_d = \frac{40}{22.2} = 1.8.$$

Going from B to A gives exactly the same number, because the averages do not depend on the direction. Quantity moves almost twice as much as price in percentage terms.

Unless a problem says otherwise, use the midpoint method whenever you are given two points. When a problem gives you the percentage changes directly (“the price rises 10 percent”), just divide them.

5.2.3 The vocabulary

Elasticity	Demand is	Meaning
$E_d = 0$	perfectly inelastic	quantity does not respond at all (vertical curve)
$E_d < 1$	inelastic	quantity moves by a smaller percentage than price
$E_d = 1$	unit elastic	quantity and price move by the same percentage
$E_d > 1$	elastic	quantity moves by a larger percentage than price
$E_d = \infty$	perfectly elastic	any higher price wipes out demand (horizontal curve)

Table 2: The range of price elasticities of demand.

The dividing line is 1, and it is the line that decides what happens to revenue when the price changes (Section 5.3). Figure 4 shows the five cases. Drawn on the same axes and the same scale, a flatter demand curve is a more elastic one.

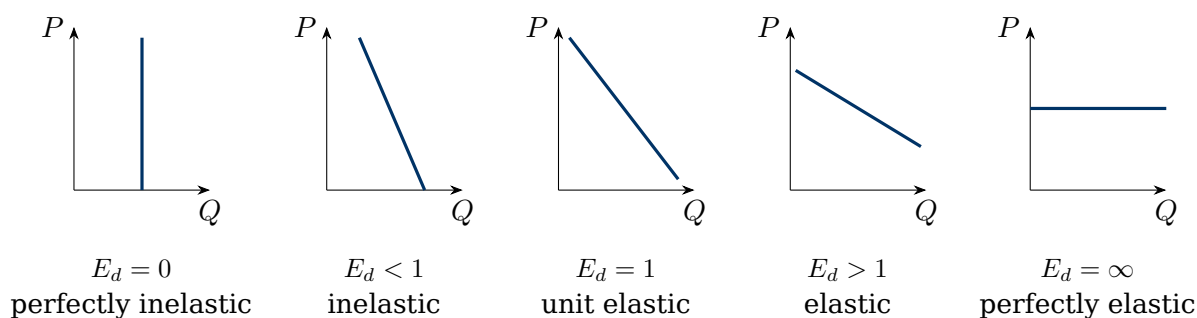


Figure 4: From perfectly inelastic to perfectly elastic. (The middle panel is only a picture: a straight line does not have one elasticity, as Section 5.2.5 shows.)

The extremes are useful reference points. Insulin for a diabetic this week is close to perfectly inelastic, and so is the demand facing a Kerala boat for the fish already in its hold. At the other end, a single wheat farmer faces demand that is close to perfectly elastic: the market price is set by thousands of sellers, and if one farmer charges a cent more, nobody buys from him.

Determinant	Demand is more elastic when	Example
Close substitutes	good substitutes exist	butter (margarine)
Necessity or luxury	the good is a luxury	sailboats, not insulin
Market definition	the market is defined narrowly	Honda Civics, not cars
Time horizon	buyers have longer to adjust	gasoline: five years, not a week

Table 3: The four determinants of the price elasticity of demand.

5.2.4 What determines the price elasticity of demand

All four come back to one question: how easily can buyers stop buying this particular thing? *Substitutes* give them somewhere to go. *Necessities* are things they will not give up. The *definition of the market* matters because narrow markets have more substitutes than broad ones: “food” has no substitute, while “vanilla ice cream” has dozens. The same logic explains why one seller’s demand is elastic even when the industry’s is not. A brand is a narrow market: when Marlboro raises its price, smokers switch to another brand and keep smoking.

Time is the determinant students most often forget. If the price of gasoline doubles, this week you still drive to work. Over five years you can buy a more efficient car, move closer to work, or change jobs. The same good has a much more elastic demand over a long horizon.

In class

Which has the more elastic demand, and why?

- (a) Insulin, or vanilla ice cream
- (b) Gasoline over the next week, or gasoline over the next five years
- (c) Cigarettes, or Marlboros
- (d) Fish at one Kerala beach at 7:45 AM, or fish in Kerala over a month

Answers. (a) Ice cream: a luxury with many substitutes; insulin is a necessity with none. (b) Five years: time lets buyers reorganize their lives around the price. (c) Marlboros: a narrower market. (d) Fish over a month. At 7:45 the local buyers are the only buyers and unsold fish is worthless at 8:00; over a month, buyers can shift purchases across days and markets. That is why the price at one beach could swing from 0 to 9.9 in a single morning.

5.2.5 Elasticity along a linear demand curve

It is tempting to read elasticity off the slope. That is a mistake. Take the Chapter 4 demand curve, $Q_d = 50 - P$. It is a straight line, so its slope is the same everywhere. Its elasticity is not (Table 4, Figure 5).

The reason is arithmetic. At a high price and a low quantity, a 10-unit change is a large *percentage* of quantity and a \$10 change is a small percentage of price. At a

Price change	Quantity change	$\% \Delta Q$ (midpoint)	$\% \Delta P$ (midpoint)	E_d
\$40 $\rightarrow$ \$30	10 $\rightarrow$ 20	$10/15 = 66.7\%$	$10/35 = 28.6\%$	2.33
\$30 $\rightarrow$ \$20	20 $\rightarrow$ 30	$10/25 = 40.0\%$	$10/25 = 40.0\%$	1.00
\$20 $\rightarrow$ \$10	30 $\rightarrow$ 40	$10/35 = 28.6\%$	$10/15 = 66.7\%$	0.43

Table 4: Elasticity along $Q_d = 50 - P$. Each step is the same \$10 and the same 10 units, but the percentages change.

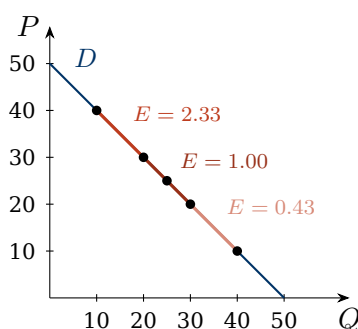


Figure 5: One slope, three elasticities. Demand is elastic at high prices and inelastic at low prices; it is unit elastic at the midpoint, $P = 25$, $Q = 25$.

low price and a large quantity, the reverse holds. Every linear demand curve is elastic above its midpoint, unit elastic at the midpoint, and inelastic below it.

Common mistake

Slope and elasticity are different things. Slope is the ratio of *changes* ($\Delta P / \Delta Q$); elasticity is the ratio of *percentage* changes. A straight line has one slope and every elasticity from zero to infinity. Never say “the” elasticity of a linear demand curve without saying where on it you are.

5.3 Elasticity and total revenue

Definition 5.4 (Total revenue). **Total revenue** is the amount buyers pay and sellers receive: $TR = P \times Q$. On a demand diagram it is the area of the rectangle below and to the left of the point on the curve.

Raise the price and two things happen at once. Every unit you still sell brings in more, which raises revenue. You sell fewer units, which lowers it. Which effect is bigger depends on whether quantity falls by a larger or a smaller percentage than price rises, which is exactly what the price elasticity measures.

Example 5.3 (Your websites, twice). In the left panel of Figure 6, demand is the one from Figure 3, with $E_d = 1.8$. Raising the price from \$200 to \$250 cuts sales from 12 to 8. Revenue falls from $200 \times 12 = \$2,400$ to $250 \times 8 = \$2,000$. Quantity fell 40 percent and price rose 22 percent: the loss (red) is bigger than the gain (blue).

In the right panel, demand is steeper: the same price rise cuts sales only from 12 to 10. By the midpoint method $\% \Delta Q = -2/11 = -18.2\%$, so $E_d = 18.2/22.2 = 0.82$. Revenue rises from \$2,400 to $250 \times 10 = \$2,500$. Now the gain is bigger than the loss.

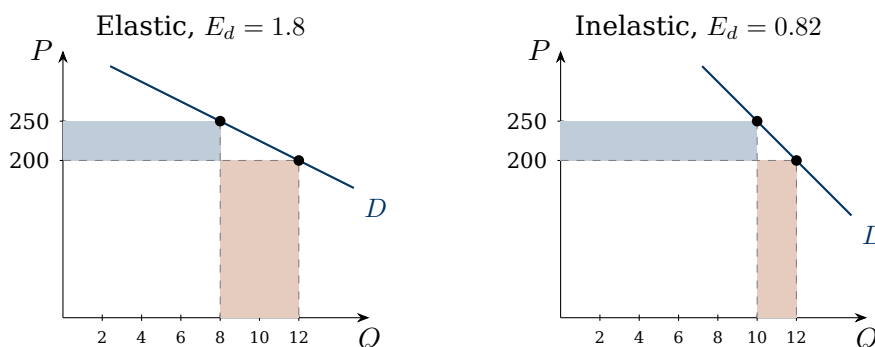


Figure 6: The same price increase, \$200 to \$250. Blue: revenue gained on the units still sold. Red: revenue lost on the units no longer sold. Left, TR falls from \$2,400 to \$2,000; right, it rises to \$2,500.

Key point

When demand is **elastic**, price and total revenue move in *opposite* directions: a price increase lowers revenue, a price cut raises it. When demand is **inelastic**, they move *together*. When demand is unit elastic, a small price change leaves revenue unchanged.

5.3.1 Revenue along a linear demand curve

Because a linear demand curve is elastic at high prices and inelastic at low ones, revenue along it first rises and then falls (Table 5, Figure 7). Revenue is largest exactly where demand is unit elastic, at the midpoint of the curve. Above the midpoint, a price cut raises revenue; below it, a price cut lowers revenue. A seller who wants more revenue should never be on the inelastic part of its demand curve: raising the price there would bring in more money from fewer customers.

P	$Q = 50 - P$	$TR = P \times Q$	
40	10	400	elastic range
30	20	600	
25	25	625	unit elastic: maximum
20	30	600	
10	40	400	inelastic range

Table 5: Total revenue along $Q_d = 50 - P$.

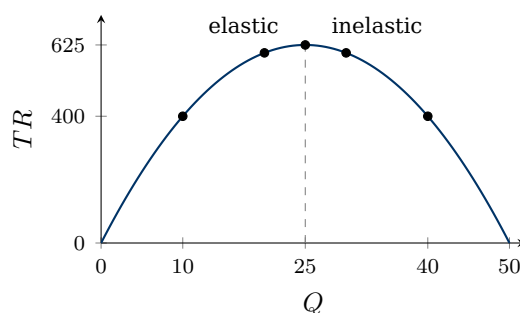


Figure 7: Total revenue $TR = Q(50 - Q)$ as quantity moves down the demand curve. It peaks at $Q = 25$, where $E_d = 1$.

In class

- Pharmacies raise the price of insulin by 10 percent. Does total spending on insulin rise or fall?
- A fare war cuts the price of a luxury cruise by 20 percent. Does the cruise lines' revenue rise or fall?

Answers. (a) Demand for insulin is inelastic, so quantity falls by less than 10 percent and spending rises. Buyers are worse off and sellers take in more. (b) A luxury cruise has substitutes and is not a necessity: demand is elastic. Quantity rises by more than 20 percent, so revenue rises. One rule, two different numbers, opposite answers.

5.4 Other demand elasticities

The same idea applies to any determinant of demand. Two are used constantly.

Definition 5.5 (Income elasticity of demand).

$$E_I = \frac{\text{percentage change in quantity demanded}}{\text{percentage change in income}}.$$

For a **normal** good the income elasticity is positive; for an **inferior** good (bus rides, instant noodles) it is negative. Among normal goods, necessities such as food and fuel have small positive income elasticities, and luxuries such as flights and jewelry have large ones.

Definition 5.6 (Cross-price elasticity of demand).

$$E_{xy} = \frac{\text{percentage change in quantity demanded of good } x}{\text{percentage change in the price of good } y}.$$

A positive cross-price elasticity means the goods are **substitutes**: when Pepsi gets more expensive, people buy more Coke. A negative one means they are **complements**: when printers get more expensive, people buy less ink. A value near zero means the goods are unrelated. These are the Chapter 4 demand shifters, now with a number attached.

Common mistake

Drop the minus sign only for the *own-price* elasticity of demand, where it is always negative. For income and cross-price elasticities, the sign is the information: it tells you whether the good is normal or inferior, and whether two goods are substitutes or complements.

5.5 The price elasticity of supply

Definition 5.7 (Price elasticity of supply).

$$E_s = \frac{\text{percentage change in quantity supplied}}{\text{percentage change in price}},$$

computed by the same midpoint method. It measures how much sellers respond to a better price.

Supply is **inelastic** when sellers cannot easily produce more: beachfront land, a Rembrandt, or this morning's catch once it has landed. No price makes more of them appear. Supply is **elastic** when sellers can expand output cheaply: T-shirts, books, most manufactured goods, where firms have spare capacity and can buy more inputs.

Two factors matter most.

- **Time.** Over a day, a factory's output is nearly fixed. Over a year it can add shifts; over several years, firms can build plants and new firms can enter. Supply is much more elastic in the long run than in the short run.
- **Capacity.** At low output, with idle machines and unused hours, a small price increase brings out a large increase in quantity. Near capacity, with plants running flat out, even a large price increase brings out little. So the same supply curve can be elastic at low quantities and inelastic at high ones (Figure 8).

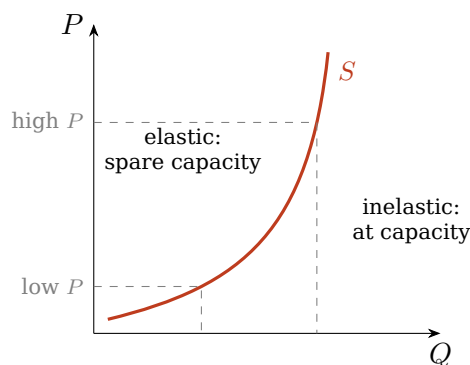


Figure 8: A supply curve that is elastic at low output and inelastic near capacity.

The Kerala beach is the extreme case of both. On the morning of the market the catch is fixed: supply is perfectly inelastic. When quantity cannot adjust, all of the adjustment lands on the price.

5.6 Three applications

5.6.1 Can good news for farming be bad news for farmers?

Suppose agronomists develop a hybrid that raises wheat yields by 20 percent, and farmers adopt it. Apply the three steps from Chapter 4. The new seed lowers the cost of producing wheat, so it shifts *supply*; it shifts it to the *right*; the price falls and the quantity rises (Figure 9).

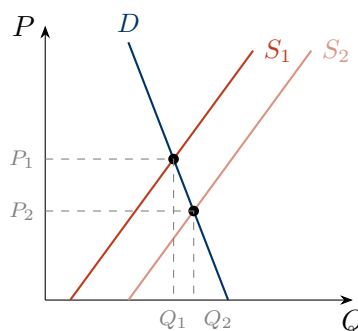


Figure 9: The wheat market after a yield-raising hybrid. Supply shifts right. With inelastic demand the price falls proportionally more than the quantity rises, so farm revenue falls.

Are farmers better off? Their revenue is $P \times Q$, and demand for staples like wheat is inelastic: nobody eats 20 percent more bread because bread got cheaper. With inelastic demand, the price falls by a larger percentage than the quantity rises, so total revenue *falls*. Each farmer is right to adopt the seed, since any one farmer who does not is left with the old yield at the new, lower price. Together, they cut their own incomes.

This is one reason the farm workforce has shrunk as farms became more productive. Agriculture employed 41 percent of the U.S. workforce in 1900 and 1.9 percent in 2000 (USDA Economic Research Service). It is also the logic behind old policies that paid farmers to leave land idle: cutting supply raises farm revenue when demand is inelastic, at the expense of consumers.

5.6.2 Why did OPEC fail to keep the price of oil high?

In October 1973 OPEC cut production. The posted price of crude oil, about \$3 a barrel before the cut, was \$11.65 by January 1974. A second round of cuts and disruptions pushed prices above \$35 a barrel by 1980. Then the price fell: during 1986 it slid from about \$26 to near \$10.

Elasticity over time explains the pattern (Figure 10). In the short run, both supply and demand for oil are inelastic. Non-OPEC producers cannot open new fields overnight, and buyers cannot quickly replace their cars and furnaces. A cut in supply therefore produces a large price increase. Over several years both curves become more elastic: new fields in the North Sea and Alaska came on line, and buyers switched to more

efficient cars and other fuels. The same cut then produces a much smaller price increase. OPEC members also had every incentive to produce more than their quotas at the high price, and many did.

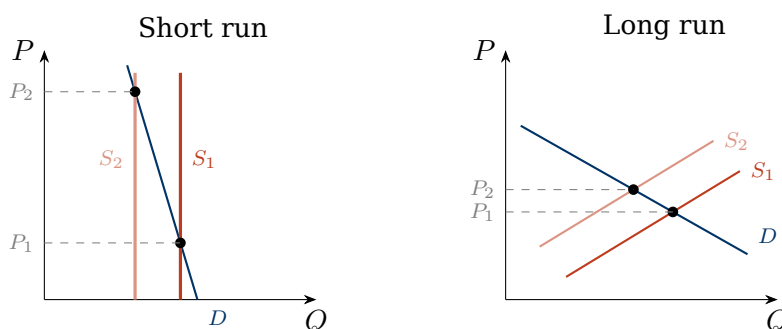


Figure 10: The same leftward shift in supply. With inelastic short-run curves (left) the price rises sharply; with elastic long-run curves (right) it barely moves.

5.6.3 Does drug interdiction increase or decrease drug-related crime?

Two policies aim to reduce the use of illegal drugs (Figure 11).

Interdiction. More federal agents seize more drugs at the border. That raises the cost of bringing drugs to market, so supply shifts left: the price rises and the quantity falls. But the demand for drugs, especially among addicted users, is inelastic. The price rises by a larger percentage than the quantity falls, so total spending on drugs *rises*. If users pay for drugs through crime, interdiction can increase drug-related crime, at least in the short run.

Treatment and education. Treatment centers and prevention programs reduce the number of people who want drugs. Demand shifts left: the price and the quantity both fall, and total spending on drugs falls with them.

Both policies reduce the quantity consumed. Only the second reduces the sellers' revenue, which is what finances the trade and what some users steal to pay for. The argument depends entirely on demand being inelastic. If it were elastic, interdiction would reduce spending too. Over a longer horizon, demand for drugs is likely more elastic, because higher prices discourage new users from starting, so the short-run and long-run effects of interdiction can differ.

5.7 Summary

- The price elasticity of demand is the percentage change in quantity demanded divided by the percentage change in price, reported as a positive number. Use the midpoint method so that the answer does not depend on the direction of the change.
- Demand is elastic when $E_d > 1$, inelastic when $E_d < 1$, and unit elastic when $E_d = 1$. It is more elastic when close substitutes exist, when the good is a luxury, when the market is defined narrowly, and when buyers have more time to adjust.

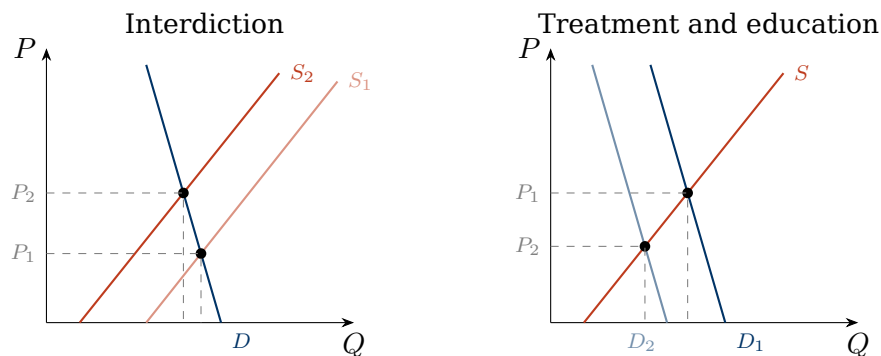


Figure 11: Left: interdiction shifts supply left (S_1 to S_2); with inelastic demand, spending $P \times Q$ rises. Right: treatment and education shift demand left; price, quantity and spending all fall.

- A linear demand curve has one slope but every elasticity: elastic above its midpoint, inelastic below it.
- Total revenue $P \times Q$ moves opposite to price when demand is elastic and with price when demand is inelastic. On a linear demand curve it is largest at the midpoint, where $E_d = 1$.
- The income elasticity is positive for normal goods and negative for inferior goods; the cross-price elasticity is positive for substitutes and negative for complements. Keep the sign for both.
- The price elasticity of supply is inelastic when output is fixed or plants are at capacity, and more elastic over longer horizons.
- When one side of a market cannot adjust its quantity, the price does all the adjusting: the Kerala beach, OPEC in the short run, farm prices after a good harvest.

Exercises

Exercise 5.1. A coffee shop raises the price of a latte from \$4 to \$5, and daily sales fall from 220 to 180.

- Compute the price elasticity of demand by the midpoint method. Is demand elastic or inelastic?
- What happens to the shop's total revenue? Is that consistent with your answer to (a)?
- Compute the elasticity instead by dividing by the starting values, first going from \$4 to \$5 and then from \$5 to \$4. What goes wrong?

Exercise 5.2. The demand for a good is $Q_d = 100 - 2P$.

- Compute total revenue at $P = 10, 20, 25, 30$, and 40. At what price is revenue largest?
- Compute the elasticity between $P = 10$ and $P = 20$, and between $P = 30$ and $P = 40$.

- (c) A seller facing this demand charges \$40 and wants more revenue. Should it raise or lower its price? What if it charges \$10?

Exercise 5.3. Studies indicate that the price elasticity of demand for cigarettes is about 0.2. A pack currently costs \$5.

- (a) The government wants to reduce smoking by 10 percent. What price is needed?
- (b) Would a tax that raises the price of cigarettes be more effective in the short run or in the long run? Explain.
- (c) Do teenagers have a higher price elasticity of demand for cigarettes than adults? Discuss using each statement: (i) it is legal for adults to drink alcohol, so many adults spend their money on alcohol rather than cigarettes; (ii) adults are more likely to be addicted to cigarettes; (iii) teenagers have less income than adults.

Exercise 5.4. Two policies are proposed to reduce the use of illegal drugs: (1) increase the number of federal agents devoted to drug interdiction; (2) open more addiction treatment centers and spend more on education. The demand for drugs is highly inelastic. Which policy is expected to reduce the total amount of money spent on drugs? Which is more likely to reduce drug-related crime? Use diagrams.

Exercise 5.5. For each case, compute the elasticity and classify the good or the pair of goods.

- (a) Incomes rise 10 percent; bus rides fall 5 percent.
- (b) Incomes rise 10 percent; restaurant meals rise 15 percent.
- (c) The price of coffee rises 20 percent; tea sales rise 8 percent.
- (d) The price of printers rises 10 percent; sales of ink cartridges fall 6 percent.

Exercise 5.6. The price of hand-knit sweaters rises from \$40 to \$60, and the quantity supplied rises from 100 to 140 a month. Compute the price elasticity of supply by the midpoint method. Would you expect it to be larger or smaller if you measured the response over three years instead of one month? Why?

Exercise 5.7. Using a supply-and-demand diagram for a single Kerala beach market on a single morning, explain (a) how the price at Badagara could be zero while markets 15 kilometers away paid 9.9 rupees per kilogram, and (b) how mobile phones could raise fishermen's profits and lower the price consumers paid at the same time.

Answers to exercises

Exercise 5.1. (a) $\% \Delta Q = (180 - 220)/200 = -20\%$ and $\% \Delta P = (5 - 4)/4.5 = 22.2\%$, so $E_d = 20/22.2 = 0.90$: inelastic. (b) Revenue rises from $4 \times 220 = \$880$ to $5 \times 180 = \$900$. With inelastic demand, price and revenue move together, so this is consistent. (c) From \$4 to \$5: $\% \Delta Q = -40/220 = -18.2\%$ and $\% \Delta P = 1/4 = 25\%$, so $E_d = 0.73$. From \$5 to \$4: $\% \Delta Q = 40/180 = 22.2\%$ and $\% \Delta P = -1/5 = -20\%$, so $E_d = 1.11$. The two answers fall on opposite sides of 1: depending on the direction, the naive method says demand is inelastic or elastic. The midpoint method removes that problem.

Exercise 5.2. (a) $Q = 80, 60, 50, 40, 20$, so $TR = 800, 1,200, 1,250, 1,200, 800$. Revenue is largest at $P = 25$, $Q = 50$, the midpoint of the demand curve. (b) Between \$10 and \$20: $\% \Delta Q = -20/70 = -28.6\%$ and $\% \Delta P = 10/15 = 66.7\%$, so $E_d = 0.43$ (inelastic). Between \$30 and \$40: $\% \Delta Q = -20/30 = -66.7\%$ and $\% \Delta P = 10/35 = 28.6\%$, so $E_d = 2.33$ (elastic). (c) At \$40 demand is elastic, so the seller should *lower* its price: revenue rises toward \$1,250. At \$10 demand is inelastic, so it should *raise* its price.

Exercise 5.3. (a) $0.2 = 10/\% \Delta P$, so $\% \Delta P = 50\%$: the price must rise to about \$7.50. Because demand is inelastic, a small reduction in quantity requires a large increase in price. (b) In the long run. Smokers need time to quit, and a higher price also keeps young people from starting. Demand is more elastic over a longer horizon, so the same tax cuts smoking by more over ten years than over one. (c) Statement (i) points the other way: adults have a legal substitute that teenagers do not, which on its own would make adults' demand *more* elastic. Statements (ii) and (iii) both make teenagers' demand more elastic: they are less likely to be addicted, and cigarettes take a larger share of a smaller income. On balance teenagers are expected to be more price-sensitive, and most studies find that they are. That is one reason cigarette taxes are defended as a way to reduce youth smoking.

Exercise 5.4. Interdiction shifts supply left. The price rises and the quantity falls, but with inelastic demand the price rises by a larger percentage, so total spending on drugs rises. Treatment and education shift demand left; the price and the quantity both fall, and spending falls. Both reduce drug use; only treatment and education reduce spending. If users finance drugs through crime, interdiction can raise drug-related crime in the short run, while treatment and education reduce it. (See Figure 11.)

Exercise 5.5. (a) $E_I = -5/10 = -0.5$: an inferior good. (b) $E_I = 15/10 = 1.5$: a normal good, and a luxury since the elasticity exceeds 1. (c) $E_{xy} = 8/20 = 0.4$: positive, so coffee and tea are substitutes. (d) $E_{xy} = -6/10 = -0.6$: negative, so printers and ink are complements.

Exercise 5.6. $\% \Delta Q = 40/120 = 33.3\%$ and $\% \Delta P = 20/50 = 40\%$, so $E_s = 0.83$: inelastic. Over three years it should be larger. In one month the existing knitters can only work longer hours; over three years more people can learn to knit and set up shop, so quantity supplied responds more to the same price.

Exercise 5.7. (a) On one morning, supply at each beach is the catch that happened to land there: a vertical line. At Badagara more boats landed than local buyers wanted at any positive price before the market closed, so the price fell to zero and the surplus fish were dumped. At Chombala and Quilandi fewer boats landed, supply sat far to the left on a steep demand curve, and the price was high. Because fish could not be moved between beaches once landed, nothing brought the prices together. (b) With phones, boats headed for the beaches that were paying. Supply moved from the glutted markets to the short ones, and prices converged. Fish that used to be dumped were now sold, so more fish reached consumers without any extra catch: buyers paid less on average, and fishermen sold their whole catch and wasted less fuel on the wrong beach. Both sides gained because the waste was eliminated, not because one side's gain came out of the other's pocket.